Network Device Sweep & Intrusion Detection Lab

Tasks

Part 1: Security Group Setup

1. Create a group called security.

2. Add at least 3 users to the group (e.g., **alice_sec, bob_sec,**

carol_sec).

3. Ensure only members of the security group can execute the sweep

script.

Part 2: Network Sweep Script

1. Create a script named **testsweep.sh**

2. The script should:

○ Accept a subnet as an argument (e.g., **192.168.15**).

○ Sweep from .1 to .254 using ping.

○ Capture the response and extract IP addresses.

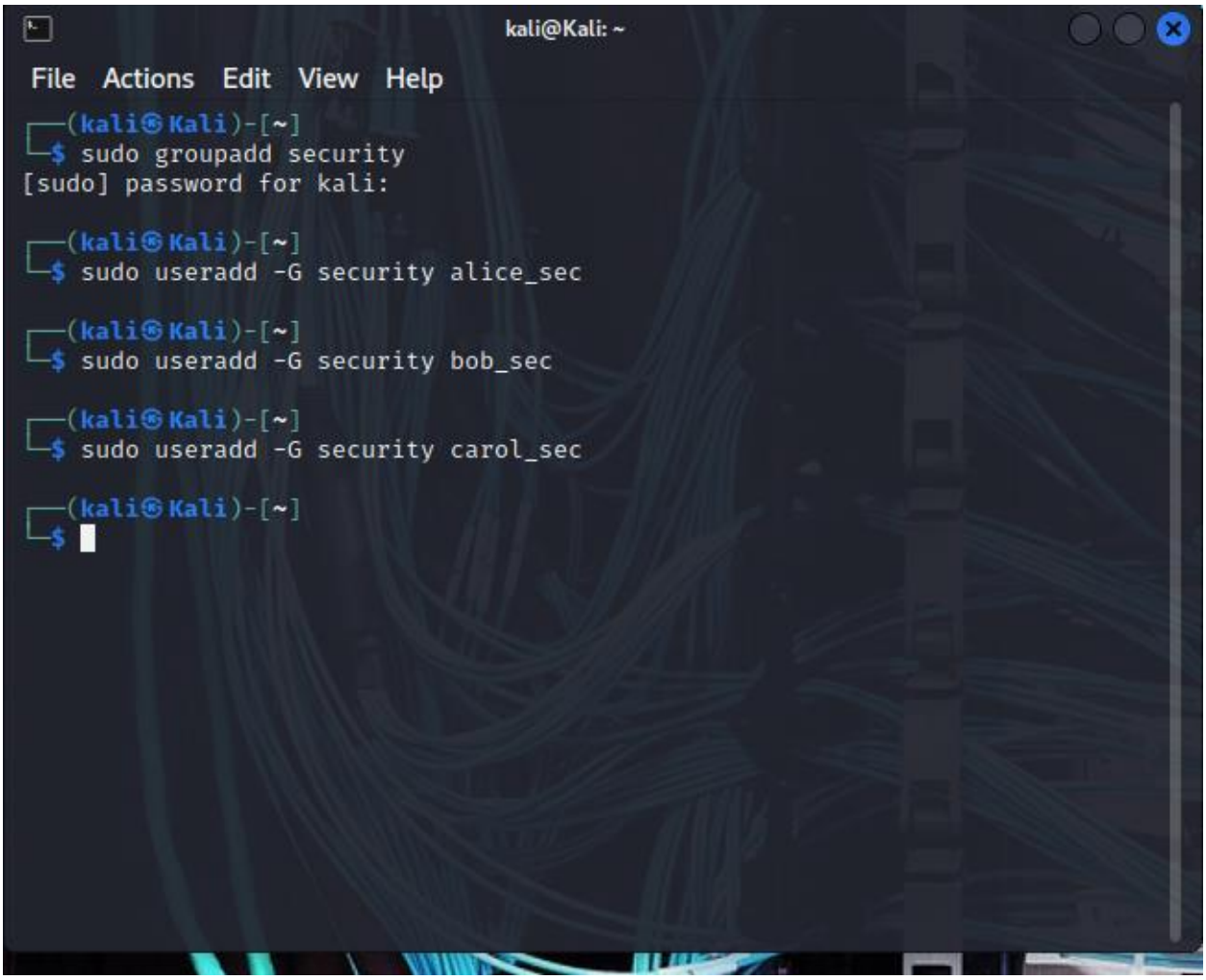
○ Write results to a file named sweep_results.txt.

Deliverables

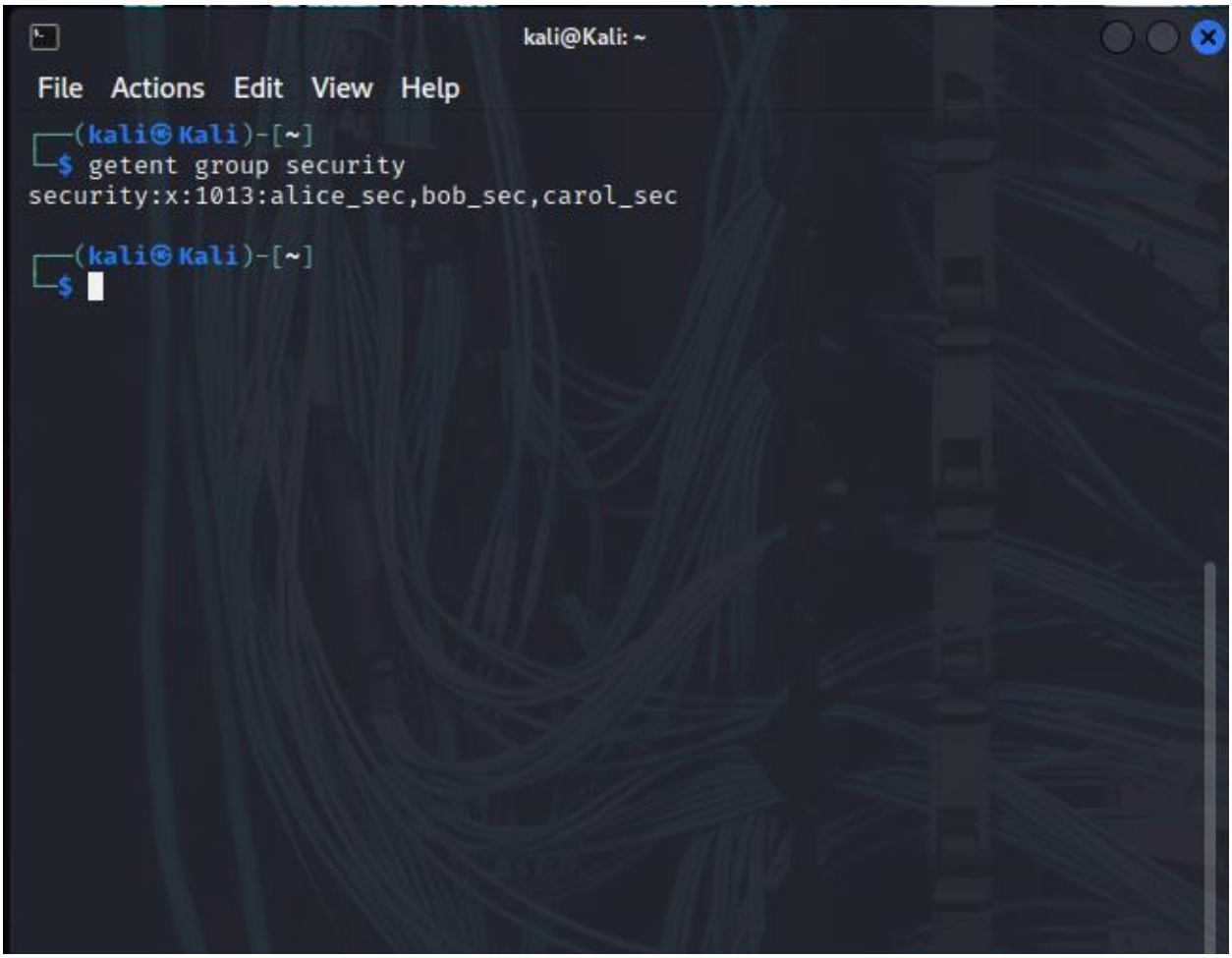
A) Below is the screenshot showing the creation

i. Creation of a group called security

ii. Creation and adding of users to the security group

B) 

B) The prompt below ‘getent group security’ shows the lists of the users created in the security group

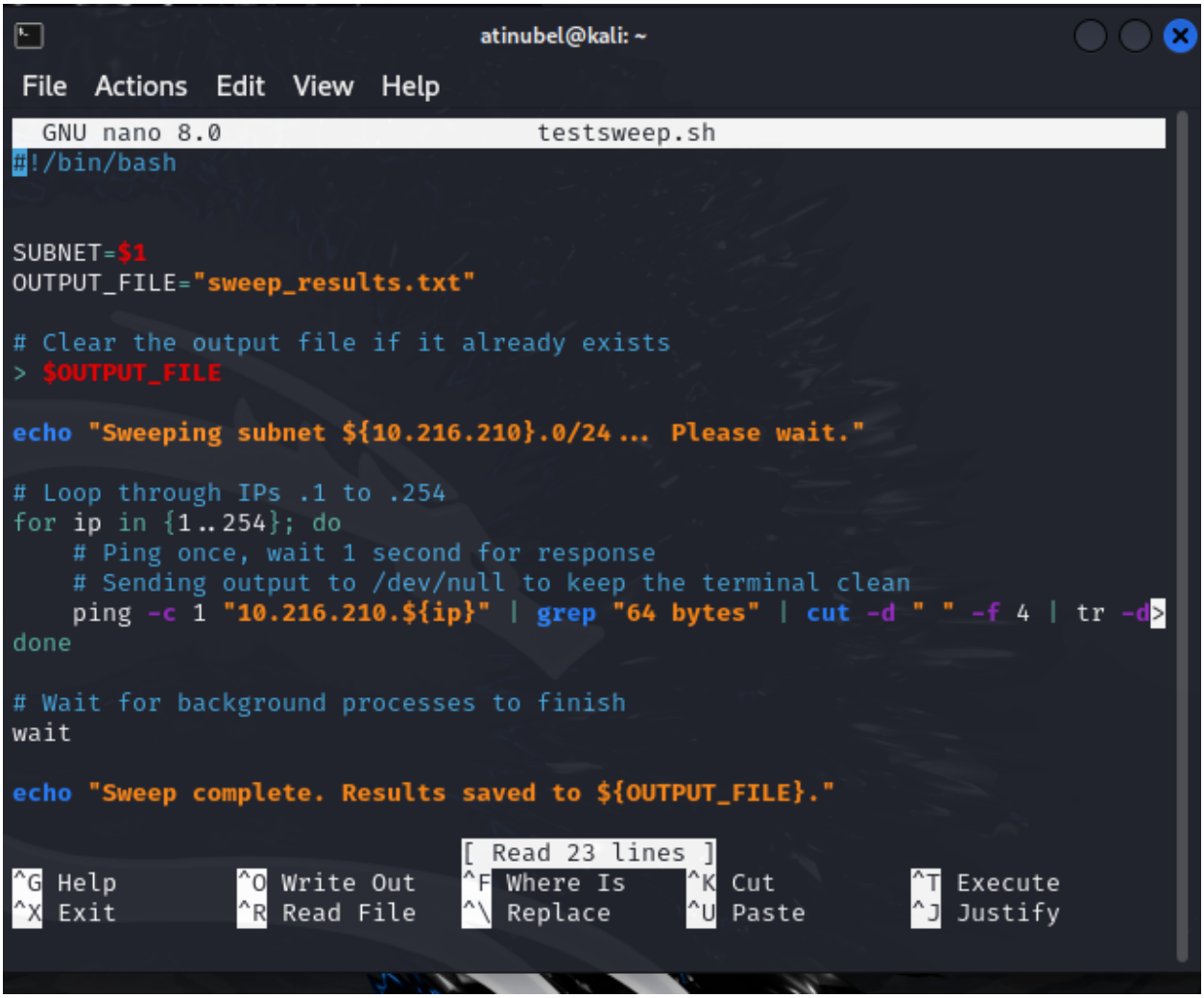


C) The below screenshot shows the

i. Creation of ‘testsweep.sh’ file

ii. The content of the ping command bash automation to list only the ip addresses in a network

iii. Saving of the result of the scan into ‘sweep_results.txt’ file



atinubel@kali: ~

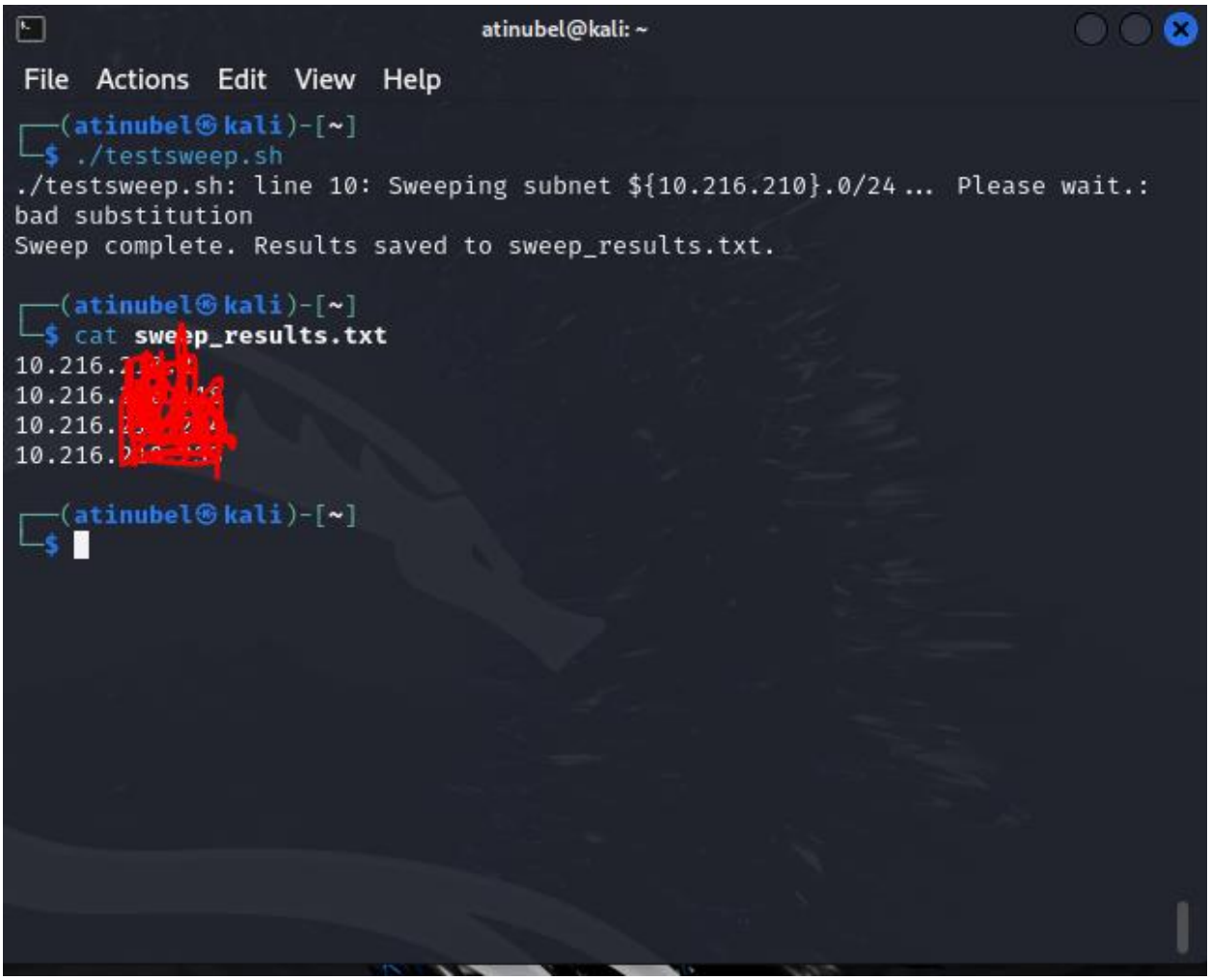
File Actions Edit View Help

```
  GNU nano 8.0                    testsweep.sh
#!/bin/bash


SUBNET=$1
OUTPUT_FILE="sweep_results.txt"

# Clear the output file if it already exists
> $OUTPUT_FILE

echo "Sweeping subnet ${10.216.210}.0/24 ... Please wait."

# Loop through IPs .1 to .254
for ip in {1..254}; do
    # Ping once, wait 1 second for response
    # Sending output to /dev/null to keep the terminal clean
< tr -d ":" >> $OUTPUT_FILE &
done

# Wait for background processes to finish
wait

echo "Sweep complete. Results saved to ${OUTPUT_FILE}."


^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

D) The file was change to an executable file using ‘chmod +x testsweep.sh’ command.

The screenshot below shows

i. The execution of the ./testsweep.sh file
ii. The result of the hosts discovered displayed in the sweep_results.txt file



E) The screenshots below show

i. The command to assign the security group exclusive permission to execute the ./testsweep.sh file

ii. The list of files in the kali directory showing the file name and permission assign to the ./testsweep.sh file

```
kali@Kali: ~
File  Actions  Edit  View  Help

┌──(kali㉿Kali)-[~]
└─$ pwd
/home/kali

┌──(kali㉿Kali)-[~]
┌──(kali㉿Kali)-[~]
└─$ sudo chmod u=rw,g=rwx,o= /home/kali/testsweep.sh
[sudo] password for kali:

┌──(kali㉿Kali)-[~]
└─$ ls -alps
total 18616
   4 drwx------ 22 kali    kali        4096 Mar  4 01:26 ./
   4 drwxr-xr-x  4 root    root        4096 Feb 19 13:13 ../
   0 -rw-------  1 kali    kali           0 Nov 29 19:16 .ICEauthority
   4 -rw-------  1 kali    kali          49 Mar  3 23:22 .Xauthority
   4 drwxr-xr-x 21 kali    kali        4096 Jan  1 21:53 .ZAP/
   4 -rw-r--r--  1 kali    kali        2303 Jan 14 15:46 .bash_history
   4 -rw-r--r--  1 kali    kali         220 Aug 14  2023 .bash_logout
   8 -rw-r--r--  1 kali    kali        5551 Aug 14  2023 .bashrc
   4 -rw-r--r--  1 kali    kali        3526 Aug 14  2023 .bashrc.original
   4 drwxr-xr-x 10 kali    kali        4096 Dec 21 19:28 .cache/
   4 drwxr-xr-x 13 kali    kali        4096 Jan 17 13:59 .config/
   4 -rw-r--r--  1 kali    kali          35 Nov 29 19:45 .dmrc
  12 -rw-r--r--  1 kali    kali       11759 Aug 14  2023 .face
   0 lrwxrwxrwx  1 kali    kali           5 Aug 14  2023 .face.icon → .face
```

kali@Kali: ~

File Actions Edit View Help

```
   0 -rw-r--r--  1 kali    kali        0 Aug 14  2023 .zsh_history
  12 -rw-r--r--  1 kali    kali    10868 Aug 14  2023 .zshrc
   4 drwxr-xr-x  2 kali    kali     4096 Dec 22 03:13 Desktop/
   4 drwxr-xr-x  2 kali    kali     4096 Nov 29 19:16 Documents/
   4 drwxr-xr-x  2 kali    kali     4096 Feb  8 18:06 Downloads/
   4 drwxr-xr-x  2 kali    kali     4096 Nov 29 19:16 Music/
   4 drwxr-xr-x  2 kali    kali     4096 Sep  3  2011 OTHER/
   4 drwxr-xr-x  2 kali    kali     4096 Mar  4 00:27 Pictures/
   4 drwxr-xr-x  2 kali    kali     4096 Nov 29 19:16 Public/
   4 drwxr-xr-x  2 kali    kali     4096 Nov 29 19:16 Templates/
   4 drwxr-xr-x  2 kali    kali     4096 Nov 29 19:16 Videos/
   4 -rw-r--r--  1 kali    kali      309 Dec 10 23:59 advent.py
   4 drwxr-xr-x  2 kali    kali     4096 Feb 19 14:36 home/
   4 -rw-r--r--  1 kali    kali       41 Dec 21 19:15 iplist.txt
18344 -rw-r--r--  1 tcpdump tcpdump 18782497 Dec 10 23:14 ladies.pcap
  24 -rw-r--r--  1 kali    kali    23323 Dec 21 19:18 paroscan.html
   0 -rwxrwx---  1 kali    itdept      0 Feb 19 14:47 project_specs.txt
   4 -rw-r--r--  1 kali    kali      198 Dec 28 00:52 pw_hashes.txt
   4 -rw-r--r--  1 kali    kali       13 Mar  4 01:10 sweep_results.txt
   4 -rw-r--r--  1 kali    kali      103 Jan 14 13:08 sxij42.txt
   4 -rw-r--r--  1 kali    kali       63 Nov 29 20:20 test2.txt
   4 -rw-r--r--  1 kali    kali       18 Nov 29 20:15 testfile.txt
   4 -rw-rwx---  1 kali    kali      243 Mar  4 01:26 testsweep.sh
   0 -rw-r--r--  1 kali    kali        0 Dec 14 20:23 virus.exe

┌──(kali㉿Kali)-[~]
└─$
```